Lean 4 Certificates for the Fermat Quintic Fourfold in $\mathbb{P}^5$: Smoothness, Jacobian-Ring Hodge Numbers $(0, 120, 580, 120)$, and the Dwork Members $\psi = 1$ (singular) and $\psi = 2$ (smooth)

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November 2026 (version 1.2.0 of the FermatQuinticHost record)

Abstract

We study the Fermat quintic fourfold $X = V(x_0^5 + \cdots + x_5^5) \subset \mathbb{P}^5$ and the Dwork-type pencil $X_\psi = V(\sum x_i^5 - 5\psi\, x_0x_1x_2x_3x_4)$ through it. In Lean 4 with Mathlib we prove: X is smooth in characteristic 0; the Jacobian ring has Hilbert series $(1+t+t^2+t^3)^6$ (length 19, total 4096) with entries $120, 580, 120$ in the Griffiths degrees $4, 9, 14$ and $h^{4,0} = 0$; the member $\psi = 1$ is singular at $[1 : 1 : 1 : 1 : 1 : 0]$; and the member $\psi = 2$ is smooth. With Griffiths' residue theorem (cited) this gives the primitive middle Hodge numbers $(0, 120, 580, 120, 0)$ for X and for X_2, checked independently against the Euler characteristic 825. Outside Lean, exact Gröbner computations show X_2 contains no plane, and character bookkeeping gives at least 240 primitive rational Hodge classes on X_2; whether they are algebraic is an open instance of the variational Hodge conjecture. Every Lean declaration depends only on the standard axioms.

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1 Introduction

The Fermat quintic fourfold is the simplest smooth quintic in $\mathbb{P}^5$ and a natural test host for formal methods in Hodge theory: its Jacobian ring is a tensor product of truncated polynomial rings, so every Hodge number is a finite count. This paper records what has been machine-checked about it, what is cited, and what remains open. Previous versions of this record (1.0.0, October 2026) contained a two-page certificate note for an Init-only file `Hodge/VF5.lean`; version 1.1.0 contained only a short update note. This version is the full paper, and supersedes both.

Contributions. (i) A Mathlib proof of smoothness of X over any field of characteristic 0 (Section 2). (ii) A decidable computation of the Jacobian-ring Hilbert series and the Griffiths entries (Sections 3–4). (iii) A Lean proof that X_1 is singular and X_2 is smooth, with an honest account of what is known about cycles on X_2 (Section 5). (iv) A complete axiom audit (Section 6).

2 Setup and smoothness

Let K be a field of characteristic 0 (for Hodge theory, $K = \mathbb{C}$) and

$$G = \sum_{i=0}^{5} x_i^5 \in K[x_0, \dots, x_5], \qquad X = V(G) \subset \mathbb{P}^5_K.$$

G is homogeneous of degree 5 (Lean: `QuinticHost.G_isHomogeneous`), so X is a hypersurface of dimension 4. Euler's identity $5G = \sum_i x_i\, \partial_i G$ (Lean: `FermatQuintic.euler_homogeneous`) shows that a common zero of the partials lies on X.

Theorem 2.1 (Lean: `QuinticHost.pderiv_G`, `QuinticHost.smooth`)**.** *$\partial_i G = 5x_i^4$, and if $\operatorname{char} K = 0$ the six partials vanish simultaneously only at $x = 0$. Consequently X is smooth (Jacobian criterion).*

Proof. $5x_i^4 = 0$ with $5 \neq 0$ in K gives $x_i^4 = 0$, hence $x_i = 0$ in the field K. □

The Jacobian criterion itself (a projective hypersurface is smooth iff the partials have no common zero off the origin) is standard and is not formalised; Lean proves the polynomial statement.

Basic invariants: $K_X = \mathcal{O}_X(5-6) = \mathcal{O}_X(-1)$, so X is Fano of index 1. By Lefschetz, $b_0 = b_2 = b_6 = b_8 = 1$ and $b_1 = b_3 = b_5 = b_7 = 0$. The topological Euler characteristic of a smooth degree-d hypersurface in $\mathbb{P}^5$ is $\chi = \frac{(1-d)^6-1}{d} + 6$; for $d = 5$ this is $\frac{4096-1}{5} + 6 = 825$, so $b_4 = 825 - 4 = 821$ and the primitive part has dimension 820.

3 The Jacobian ring and its Hilbert series

The Jacobian ideal is $J = (\partial_0 G, \dots, \partial_5 G) = (x_0^4, \dots, x_5^4)$ and the Jacobian ring is

$$R = \mathbb{C}[x_0, \dots, x_5]/J \;=\; \bigotimes_{i=0}^{5} \mathbb{C}[x_i]/(x_i^4).$$

Since $x_0^4, \dots, x_5^4$ is a regular sequence, R is a graded Artinian complete intersection with monomial basis $\{x^a : 0 \le a_i \le 3\}$, so

$$H_R(t) = \Big(\frac{1-t^4}{1-t}\Big)^6 = (1+t+t^2+t^3)^6, \qquad \dim R = 4^6 = 4096,$$

with socle in degree $6 \cdot 3 = 18$ and the symmetry $\dim R_k = \dim R_{18-k}$ (Macaulay duality). For an arbitrary smooth quintic in six variables the partials still form a regular sequence of six forms of degree 4, so the same Hilbert series holds (cited, not formalised). For example R_4 is spanned by

k	0	1	2	3	4	5	6	7	8	9
$\dim R_k$	1	6	21	56	**120**	216	336	456	546	**580**
k	10	11	12	13	14	15	16	17	18	
$\dim R_k$	546	456	336	216	**120**	56	21	6	1	

Table 1: Coefficients of $(1+t+t^2+t^3)^6$. Sum = 4096. Bold: the Griffiths degrees.

degree-4 monomials in six variables with all exponents ≤ 3: $\binom{9}{5} - 6 = 126 - 6 = 120$ (only the six x_i^4 are excluded).

4 Griffiths residues and the Hodge numbers

Theorem 4.1 (Griffiths [1], cited)**.** *For a smooth hypersurface $X = V(G) \subset \mathbb{P}^{n+1}$ of degree d, $n = \dim X$,*

$$H^{n-q,q}_{\text{prim}}(X) \;\cong\; R_{d(q+1)-(n+2)}.$$

Here $n = 4$, $d = 5$, so $h^{4-q,q}_{\text{prim}} = \dim R_{5q-1}$:

q	degree $5(q+1)-6$	$h^{4-q,q}_{\text{prim}}$	Lean
0	-1	0	`quintic_h40_zero`
1	4	120	`quintic_primitive_hodge`
2	9	580	`quintic_primitive_hodge`
3	14	120	`quintic_primitive_hodge`
4	19	0	(beyond socle degree 18)

The total $0+120+580+120+0 = 820$ agrees with the Euler-characteristic computation of Section 2, an independent check. Adding h^2 gives the middle row of the Hodge diamond

$$(h^{4,0}, h^{3,1}, h^{2,2}, h^{1,3}, h^{0,4}) = (0, 120, 581, 120, 0), \qquad b_4 = 821.$$

Remark 4.2 (What is and is not formalised)**.** Lean verifies the *counting*: the coefficient list of $(1+t+t^2+t^3)^6$ and its entries at the Griffiths degrees. The identification of $\dim R_k$ with a Hodge number is Griffiths' theorem, which is cited. In particular $h^{3,1} = 120 \neq 0$; only $h^{4,0}$ (and $h^{0,4}$) vanish.

5 The Dwork-type pencil

Consider the pencil

$$G_\psi = \sum_{i=0}^{5} x_i^5 - 5\psi\, x_0x_1x_2x_3x_4, \qquad X_\psi = V(G_\psi) \subset \mathbb{P}^5.$$

The product omits x_5, so every member is homogeneous of degree 5 (an earlier proposal using $x_0 \cdots x_5$ is *not* homogeneous and does not define a hypersurface in $\mathbb{P}^5$; this is also recorded in Lean as `FermatQuintic.G_not_euler_degree5`). X_0 is the Fermat quintic.

5.1 $\psi = 1$ is singular

Proposition 5.1 (Lean: `DeformedQuinticHost.singular_at_p`, `not_smooth`)**.** *The point $p = [1:1:1:1:1:0]$ lies on X_1 and all partial derivatives of G_1 vanish at p. Hence X_1 is singular.*

Proof. $G_1(p) = 5 - 5 = 0$; $\partial_i G_1 = 5x_i^4 - 5\prod_{j\le 4, j\neq i} x_j$ for $i \le 4$ equals $5 - 5 = 0$ at p; and $\partial_5 G_1 = 5x_5^4 = 0$. □

More generally X_ψ is singular when $\psi^5 = 1$ (the points $[\zeta^{a_0} : \cdots : \zeta^{a_4} : 0]$); only $\psi = 1$ is formalised. The Griffiths machinery of Section 4 does not apply to X_1.

5.2 $\psi = 2$ is smooth

Theorem 5.2 (Lean: `DworkQuinticPsi2.smooth`)**.** *Over a field K of characteristic 0, the partials of $G_2 = \sum x_i^5 - 10\, x_0 \cdots x_4$ vanish simultaneously only at 0. Hence X_2 is a smooth quintic fourfold.*

Proof. The partials are $5x_5^4$ and $5x_i^4 - 10\prod_{j\le 4, j\neq i} x_j$ ($i \le 4$). So $x_5 = 0$ and $x_i^4 = 2\prod_{j\neq i} x_j$. Put $P = x_0x_1x_2x_3x_4$. Multiplying the five relations, $P^4 = \prod x_i^4 = 2^5 \prod_i \prod_{j\neq i} x_j = 32P^4$, so $31P^4 = 0$ and $P = 0$. Then $x_i^5 = x_i \cdot 2\prod_{j\neq i} x_j = 2P = 0$ for each $i \le 4$. This is exactly the Lean proof (it uses `linear_combination` for each step). □

By Theorem 5.2, Section 4 applies verbatim to X_2: primitive Hodge numbers $0, 120, 580, 120, 0$ (Lean arithmetic: `dwork_primitive_hodge`). Independently, Singular computed the Hilbert function of $\mathbb{Q}[x]/J(G_2)$ as $1, 6, 21, 56, 120, 216, 336, 456, 546, 580, 546, \ldots, 1$ (total 4096), matching Table 1 (script `scripts/dwork_quintic_psi2/hilb.sing`).

5.3 Computations outside Lean for $\psi = 2$

- **No planes.** A plane in $\mathbb{P}^5$ is the row space of a 3×6 matrix in reduced form. S_5 (permuting $x_0, \ldots, x_4$) preserves G_2, so the 20 Grassmannian charts reduce to two orbit representatives (pivots $\{0,1,2\}$ and $\{0,1,5\}$). Restricting G_2 gives 21 equations in 9 unknowns; a Gröbner basis over $\mathbb{Q}$ is $\{1\}$, so X_2 contains no plane over $\mathbb{C}$. The same code finds the planes of the Fermat quintic (sanity check). (Script `planes.sing`; exact, not Lean.)
- **Hodge classes.** Let $H = \{a \in (\mathbb{Z}/5)^6 : a_0 + \cdots + a_4 \equiv 0\}$/diagonal (order 625) act by $x_i \mapsto \zeta^{a_i} x_i$; it preserves every G_ψ. Deforming to $\psi = 0$, the 820 Fermat characters group into 500 H-eigenspaces (240 of rank 1, 200 of rank 2, 60 of rank 3). The 240 rank-1 eigenspaces are, with all Galois conjugates, of type $(2,2)$, so they give *at least 240 primitive rational Hodge classes on X_ψ for every smooth ψ*, including $\psi = 2$ (script `chars.py`; bookkeeping plus cited theory). The exact rank at $\psi = 2$ is not computed.

5.4 Status of the Hodge conjecture for X_2: open

For the Fermat quintic the Hodge conjecture is known (Shioda [2], Ran [3]: prime degree), with cycles built from linear subspaces. At $\psi = 2$ those planes do not deform (X_2 contains none), and G_2 has 7 monomials in 6 variables, so it is not of Delsarte type. Showing that the 240 classes above stay algebraic along the pencil is an instance of the *variational Hodge conjecture*, which is open. We found no published result for $\psi = 2$. **No claim of the Hodge conjecture for X_2 is made.**

6 Lean formalisation

6.1 Where the proofs live

All machine-checked statements cited in this paper are in the repository `https://github.com/BenFrohman/HODGE` (Lean 4 `v4.22.0`, Mathlib `v4.22.0`), in three files:

- `Hodge/QuinticHost.lean` (Fermat quintic, merged in PR #26);
- `Hodge/DeformedQuinticHost.lean` (Dwork member $\psi = 1$, PR #26);
- `Hodge/DworkQuinticPsi2.lean` (Dwork member $\psi = 2$, PR #28).

The counting functions `conv`, `jacobianSeries` and `primHodge` are defined in `Hodge/FermatCubicHost.lean` and reused. The polynomial is an honest `MvPolynomial (Fin 6) K` over an arbitrary field `K`; partial derivatives are Mathlib's `MvPolynomial.pderiv`. None of the files uses `sorry`, `native_decide` or a user-declared `axiom`; the repository CI enforces this with a source scan.

6.2 Theorem inventory

Lean declaration	Statement
`QuinticHost.G_isHomogeneous`	$G = \sum_{i=0}^{5} x_i^5$ is homogeneous of degree 5.
`QuinticHost.pderiv_G`	$\partial G/\partial x_i = 5x_i^4$.
`QuinticHost.smooth`	If $\operatorname{char} K = 0$ and all $\partial_i G(x) = 0$ then $x = 0$.
`QuinticHost.quintic_series`	The coefficient list of $(1+t+t^2+t^3)^6$ has length 19 and sum 4096.
`QuinticHost.quintic_primitive_hodge`	$R_4 = 120$, $R_9 = 580$, $R_{14} = 120$.
`QuinticHost.quintic_h40_zero`	$5 \cdot 1 < 6$ (so the Griffiths degree for $h^{4,0}$ is negative).
`QuinticHost.quintic_h31_ne_zero`	$R_4 \neq 0$.
`DeformedQuinticHost.G_isHomogeneous`	$G_1 = \sum x_i^5 - 5x_0x_1x_2x_3x_4$ is homogeneous of degree 5.
`DeformedQuinticHost.singular_at_p`	$G_1(p) = 0$ and $\partial_i G_1(p) = 0$ for all i, $p = (1,1,1,1,1,0)$.
`DeformedQuinticHost.not_smooth`	The Jacobian criterion fails for G_1.
`DworkQuinticPsi2.G_isHomogeneous`	$G_2 = \sum x_i^5 - 10x_0x_1x_2x_3x_4$ is homogeneous of degree 5.
`DworkQuinticPsi2.eval_pderiv`	Explicit formulas for the six partials of G_2.
`DworkQuinticPsi2.smooth`	If $\operatorname{char} K = 0$ and all $\partial_i G_2(x) = 0$ then $x = 0$.
`DworkQuinticPsi2.psi_not_root`	$2^5 \neq 1$.
`DworkQuinticPsi2.dwork_primitive_hodge`	$R_4 = 120$, $R_9 = 580$, $R_{14} = 120$ for the series $(1+t+t^2+t^3)^6$.

6.3 Axiom audit

The HODGE CI runs `lake env lean AxiomAudit.lean` (one `#print axioms` per declaration) and fails unless every declaration depends only on Lean's standard axioms `propext`, `Classical.choice`, `Quot.sound`, or on none. Output on `main` (the $\psi = 2$ declarations were added to the audit in HODGE PR #31):

Declarations	`#print axioms`
`QuinticHost.{G_isHomogeneous, pderiv_G, smooth}`	`[propext, Classical.choice, Quot.sound]`
`QuinticHost.{quintic_series, quintic_primitive_hodge, quintic_h31_ne_zero}`	`[propext]`
`QuinticHost.quintic_h40_zero`	none
`DeformedQuinticHost.*` (3 declarations)	`[propext, Classical.choice, Quot.sound]`
`DworkQuinticPsi2.{G_isHomogeneous, eval_pderiv, smooth}`	`[propext, Classical.choice, Quot.sound]`
`DworkQuinticPsi2.{psi_not_root, dwork_primitive_hodge}`	`[propext]`

`Classical.choice` enters through Mathlib's polynomial and field library, not through any step specific to this paper. The counting theorems are closed by `decide` and use only `propext`.

7 The standalone certificate repository

The repository `https://github.com/BenFrohman/FermatQuinticHost` contains the original Init-only Lean 4.22 file `Hodge/VF5.lean` (no Mathlib), which works over $\mathbb{Z}$ with explicit partials $\partial_i F_5 = 5x_i^4$ and proves:

- `euler_F5`: Euler's identity $5F_5 = \sum x_i \partial_i F_5$ over $\mathbb{Z}$;
- `affine_cone_isolated_at_origin`: the partials vanish together only at $0 \in \mathbb{Z}^6$ (via `int_five_mul_prod4_eq_zero` and `partial_as_prod`);
- `G_euler_defect`: the non-homogeneous expression $F_5 - 5x_0 \cdots x_5$ fails Euler's identity in degree 5, i.e. it is not a quintic form;
- `h31_dim`: the arithmetic $126 - 6 = 120$ behind $\dim R_4$;
- bookkeeping: `host_is_VF5`, `host_dim`, and `sextic_Z_status_uncomputed` (a flag stating that the algebraic part Z on the related sextic is not computed there).

All 10 audited declarations depend only on the standard axioms (CI: `AxiomAudit.lean`). Over $\mathbb{Z}$ the isolation statement is weaker than smoothness over $\mathbb{C}$; the Mathlib proof in HODGE (`QuinticHost.smooth`) is the one used in this paper.

8 What is proved, computed, and cited

Statement	Status
X smooth (char. 0); $\partial_i G = 5x_i^4$	Lean
Hilbert series $(1+t+t^2+t^3)^6$; entries 120, 580, 120; $h^{4,0}$ degree negative	Lean (`decide`)
$R \cong \bigotimes \mathbb{C}[x_i]/(x_i^4)$ has that Hilbert series	elementary; series computed in Lean
Same series for every smooth quintic	cited (regular sequence)
$h^{4-q,q}_{\text{prim}} = \dim R_{5q-1}$	cited (Griffiths)
$\chi(X) = 825$	standard formula
X_1 singular at $[1:1:1:1:1:0]$	Lean
X_2 smooth	Lean
X_2 Jacobian Hilbert function	Singular (exact over $\mathbb{Q}$)
X_2 contains no plane	Singular (Gröbner basis $\{1\}$ over $\mathbb{Q}$)
≥ 240 primitive rational Hodge classes on X_2	Python bookkeeping + cited theory
Hodge conjecture for X_0	cited (Shioda, Ran)
Hodge conjecture for X_2	**open**

9 Corrections relative to earlier versions

- The Init-only certificate (1.0.0) is retained but is not a smoothness proof over $\mathbb{C}$; it is now supplemented by the Mathlib proof.
- Version 1.1.0 stated that smoothness of members with $\psi^5 \neq 1$ was not proven; $\psi = 2$ is now proven (HODGE PR #28).
- The expression $\sum x_i^5 - 5x_0 \cdots x_5$ that appeared in early discussions is not homogeneous and is not used.

10 Reproducibility

- Build: `git clone https://github.com/BenFrohman/HODGE; cd HODGE; lake exe cache get; lake build`.
- Axiom audit: `lake env lean AxiomAudit.lean`; CI workflow `Lean certificate`.
- Singular scripts for $\psi = 2$: `scripts/dwork_quintic_psi2/` (`hilb.sing`, `planes.sing`, `fermat.sing`, `chars.py`) with outputs committed next to them.
- Standalone certificate repository: `https://github.com/BenFrohman/FermatQuinticHost`, which carries its own CI with the same axiom check.

Scope

This paper proves smoothness statements and Jacobian-ring counts in Lean and cites Griffiths' theorem for the Hodge-theoretic interpretation. It makes no claim about algebraic cycles on the quintic fourfold beyond the cited Fermat results, and it does not claim the Hodge conjecture for any non-Fermat member of the pencil.

References

[1] P. A. Griffiths, On the periods of certain rational integrals I, II, *Ann. of Math.* 90 (1969), 460–541.

[2] T. Shioda, The Hodge conjecture for Fermat varieties, *Math. Ann.* 245 (1979), 175–184.

[3] Z. Ran, Cycles on Fermat hypersurfaces, *Compositio Math.* 42 (1980), 121–142.

[4] C. Voisin, *Hodge Theory and Complex Algebraic Geometry I, II*, Cambridge Univ. Press, 2002–2003.

[5] The mathlib Community, The Lean mathematical library, *CPP 2020*, 367–381.